# Do LLM Self-Explanations Tell One Story? A Chance-, Ceiling-, and Reliability-Corrected Audit of Cross-Paradigm Agreement

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## Abstract

Large language models are routinely asked to explain their own predictions through interchangeable-looking devices: token highlighting, free-text rationales, counterfactual rewrites, and importance rankings. We ask whether these elicitation paradigms actually name the same evidence, in a pre-registered study of 3 model families (two open-weight, one proprietary) $\times$ 4 datasets (2,400 instances), including the shortcut-resistant CAD-IMDb. Because raw token overlap is confounded by chance and by set-size geometry, we introduce $\mathrm{ECS}_{\mathrm{adj}}$: an exact-hypergeometric chance correction combined with a geometric ceiling correction and paradigm-balanced aggregation; and because the elicitation instruments are themselves unreliable, we measure per-strategy self-consistency ceilings under prompt paraphrase and apply classical reliability disattenuation. Agreement exceeds chance in every model$\times$dataset cell (12/12, Holm $p \leq 0.0012$; pooled $\mathrm{ECS}_{\mathrm{adj}} = 0.43$ on a 0=chance, 1=ceiling scale), but is far from uniform: after reliability correction, extraction$\leftrightarrow$counterfactual divergence largely disappears (corrected agreement $0.88$–$0.96$), whereas rationale-involving divergence persists (rationale–counterfactual $0.39$, 95% CI $[0.35, 0.43]$). Two models explaining the same input with the same paradigm agree more than one model explaining it with two paradigms, on all four datasets. Consensus tokens are causally load-bearing—erasing them flips predictions $2.0$–$2.7\times$ more than matched random controls—yet a pre-registered test finds agreement does *not* predict simulatability under explanation-independent perturbations. Explanation paradigms are not interchangeable views of one underlying story, and internal agreement cannot be assumed to certify downstream utility.

## 1 Introduction

When practitioners want to know *why* an LLM made a prediction, they ask it—and the asking can take several standard forms: score the importance of each token (Huang et al., 2023), give a one-sentence rationale (Camburu et al., 2018; Wiegreffe et al., 2021), produce a minimal edit that would flip the label (Ross et al., 2021), or rank the most important words. These paradigms are used interchangeably in practice, as if they were different renderings of one underlying explanation. For post-hoc explainers of classical models, the "disagreement problem" (Krishna et al., 2022) showed that different attribution methods routinely name different evidence. Whether the same fragmentation afflicts the *self*-explanations of instruction-tuned LLMs—one model, one prediction, four ways of asking—is the question of this paper.

Answering it is mostly a measurement problem, with three obstacles. First, *chance*: two random token subsets of a short input overlap substantially, so raw Jaccard rewards brevity. Second, *geometry*: a counterfactual edit touches 1–2 tokens while a salience list contains 3–10, capping achievable Jaccard far below 1 for some pairs but not others, so a flat average measures set-size geometry more than agreement. Third, *reliability*: the instruments are themselves noisy—under mild prompt paraphrase, per-strategy self-agreement ranges from $0.62$ to $0.90$—so a low observed value is ambiguous between *real paradigm divergence* and *unreliable elicitation*.

We address all three. Our estimator, $\mathrm{ECS}_{\mathrm{adj}}$, normalizes each pairwise Jaccard by its exact hypergeometric expectation (chance floor) and its size-determined maximum (ceiling), then aggregates the five cross-paradigm pairs into three paradigm-level contrasts with equal weight. Per-strategy test–retest reliabilities, measured with a paraphrase ablation on every model, feed a classical Spearman disattenuation (Spearman, 1904): if the instruments were perfectly reliable, how much would the paradigms agree? All estimands, test families, exclusion floors, and amendments were pre-registered before collection—including a fourth, counterfactually-revised dataset (Kaushik et al., 2020) that blunts the "near-solved benchmark" objection, and a simulatability bridge (Chen et al., 2024) whose result we report as the null it produced.

**Findings.** On 2,400 instances (3 models $\times$ 4 datasets $\times$ 200), elicited at temperature 0 through frozen, literature-anchored prompts:

1. **Above-chance agreement everywhere.** Complete-case $\text{ECS}_{\text{adj}}$ is positive in all 12 model$\times$dataset cells (range 0.28–0.62; Holm-corrected sign-flip $p \leq 0.0012$ in every cell), pooled $0.43$—roughly $40\%$ of the way from chance to the geometric ceiling (§5.1).

2. **Agreement is structured by paradigm distance, and reliability correction sharpens the structure.** Observed agreement orders extraction–perturbation $>$ extraction–rationalization $>$ rationalization–perturbation (complete-case within-instance means $0.615 > 0.395 > 0.286$). After disattenuation, extraction$\leftrightarrow$counterfactual pairs approach the ceiling (H–CF $0.88$, RO–CF $0.96$)—their observed divergence is largely elicitation noise—while every rationale-involving pair stays far below 1 (R–CF $0.39$, CI $[0.35, 0.43]$): free-text rationales genuinely select different evidence (§5.2).

3. **Explanations are paradigm-shaped more than model-shaped.** Two *different* models explaining the same input with the same strategy agree more than *one* model explaining it with two strategies, on all 4 datasets (matched paired contrast, all CIs above 0; §5.3).

4. **Consensus is causally load-bearing but not decision-useful.** Erasing the tokens that $\geq$3 strategies agree on flips the model's own prediction 2.0–2.7$\times$ more often than occurrence-matched random controls (all models, both operators, $p_{\text{Holm}}$=0.0002; §5.4). Yet agreement does not predict simulatability under explanation-independent perturbations: explanations barely improve a simulator over a no-explanation baseline (pooled gains $-0.015$ to $+0.016$), and the pre-registered association between $\text{ECS}_{\text{adj}}$ and simulatability gain is null (§5.5).

We release the full pipeline: frozen prompts with hashes, curated datasets with datasheets, per-instance artifacts, seeds, and the pre-registration with dated amendments.

## 2 Related Work

**Faithfulness and self-consistency.** Interpretability work distinguishes an explanation being *plausible* from being *faithful* to the model's computation (Jacovi and Goldberg, 2020; Lyu et al., 2024). Turpin et al. (2023) showed chain-of-thought can misstate the true drivers of predictions, and Madsen et al. (2024) found self-explanation faithfulness is model- and task-dependent. Parcalabescu and Frank (2024) argue most existing tests—including erasure-style checks—measure *self-consistency* rather than faithfulness. We adopt that framing: all our axes are consistency measures, and none certifies faithfulness (§7).

**Agreement between explanation methods.** The disagreement problem (Krishna et al., 2022) quantified how post-hoc attribution methods conflict for classical models; for LLMs, Huang et al. (2023) compared self-generated saliency to LIME/occlusion and Randl et al. (2024) report related single-model reliability results. We differ in three ways: we compare four *self*-elicitation paradigms (not model-external attributions) across three model families and four datasets; we correct for both chance and set-size ceiling with an exact null; and we read observed agreement against measured instrument reliability—to our knowledge the first application of psychometric disattenuation to explanation agreement.

**Counterfactuals and simulatability.** Minimal contrastive editing (Ross et al., 2021) defines the counterfactual construct we elicit, and Dehghanighobadi et al. (2025) study LLM self-counterfactuals in isolation. Kommiya Mothilal et al. (2021) formalize why attribution (sufficiency-oriented) and counterfactual (necessity-oriented) evidence diverge by construction; our rationalization–perturbation results quantify that divergence once chance, ceiling, and reliability are controlled. Counterfactual simulatability (Chen et al., 2024) operationalizes an explanation's practical value—does it help an observer predict model behavior on perturbed inputs?—and we pre-registered a bridge from agreement to a strategy-fair variant of it (§5.5). For verbalized confidence we replicate the over-confidence clustering of Xiong et al. (2024) and Tian et al. (2023).

## 3 Measuring Cross-Paradigm Agreement

### 3.1 Four elicitation paradigms, one token space

For each instance the model first classifies the text; every explanation is then elicited for the model's *own* predicted label (a label-conditioned, post-hoc protocol; see §7 for what this implies). The four strategies are: **H** (highlighting): score every word 1–10 for importance (Huang et al., 2023); the evidence set is a length-proportional top-$k$ ($k = \max(3, \mathrm{round}(L/5))$ content words, $L$ = content-word count), with repeated words aggregated by max score and score ties broken alphabetically in the normalized token space for determinism. The $k$ rule follows Huang et al. (2023)'s length-proportional selection and was fixed in the pre-registered plan before collection. Because $k$ is a design choice with no canonical value, we also report a size-robust composite built from overlap coefficients rather than Jaccard, which preserves the paradigm ordering (E–P $0.786 >$ E–R $0.689 >$ R–P $0.526$ pooled); a fully $k$-free graded-weight variant is left to future work. **R** (rationale): a one-sentence free-text explanation (Wiegreffe et al., 2021); content words are extracted with a POS filter. **CF** (counterfactual): a minimal edit that flips the label, constrained to $\leq \frac{1}{3}$ of words (Ross et al., 2021); the evidence set is the tokens changed, and validity requires the model's verified label flip. **RO** (rank ordering): the 5 most important words, ranked. Three paradigms group these four strategies: *extraction* (E) = {H, RO}, *rationalization* (R) = {R}, *perturbation* (P) = {CF}. In pair names (H–R) letters denote strategies; in component names (E–R, R–P) they denote paradigms. All evidence sets are mapped into one shared token space (lowercasing, punctuation stripping, stopword removal, lemmatization), so "terrible" and "Terrible." match across strategies. The H–RO pair is same-paradigm and serves as a within-paradigm reference; it is excluded from the cross-paradigm composite.

### 3.2 The $\mathrm{ECS}_{\mathrm{adj}}$ estimator

For evidence sets $A, B$ drawn from an instance vocabulary of size $V$, raw Jaccard $J$ has two structural confounds. Under uniform random selection the intersection is hypergeometric, giving the *exact* chance expectation

$$\mathbb{E}[J] = \sum_k \binom{a}{k}\binom{V-a}{b-k}\binom{V}{b}^{-1} \frac{k}{a+b-k}, \tag{1}$$

with $a{=}|A|, b{=}|B|$ (no Monte-Carlo error, no seed). And the best achievable value is the nesting ceiling $J_{\max} = \min(a,b)/\max(a,b)$: a *perfect* 1-token CF against a 5-token salience set caps at $J = 0.2$. We therefore score every pair with the adjusted Jaccard

$$\mathrm{AJ}(A,B) = \frac{J - \mathbb{E}[J]}{J_{\max} - \mathbb{E}[J]}, \tag{2}$$

a $\kappa$-style normalization: $0$ = chance, $1$ = maximum agreement achievable given the sizes, comparable across pairs, instances, and datasets regardless of geometry. Pairs with $J_{\max} - \mathbb{E}[J] < \varepsilon$ (pre-registered $\varepsilon = 0.10$) are degenerate—chance and ceiling nearly coincide—and are excluded with a counted flag rather than allowed to dominate the mean. AJ is not bounded below by $-1$: its floor is $(J_{\min} - \mathbb{E}[J])/(J_{\max} - \mathbb{E}[J])$, which equals $-\mathbb{E}[J]/(J_{\max} - \mathbb{E}[J])$ in the usual case but is higher when $a + b > V$ forces a non-empty intersection ($J_{\min} = (a + b - V)/V$); $5.8\%$ of scored pairs are in this forced-overlap regime, and the $\varepsilon$ guard does not exclude them. The exact null already integrates over this constraint (the hypergeometric sum starts at $k = \max(0, a + b - V)$), so the estimand is unaffected—only the floor is instance-dependent. Because the resulting null is left-skewed (bounded above at $+1$, long negative tail), a one-sided sign-flip test for $\mathrm{mean} > 0$ rejects less often than a symmetric-null test would at the same effect size: the permutation distribution inherits the same skew, so extreme negative draws inflate its upper tail and raise the rejection threshold. Negative cell means are therefore reported only as "not above chance," never as signed below-chance effects.

The five cross-paradigm pairs aggregate into three paradigm contrasts with equal weight,

$$\begin{aligned}
\text{E–R} &= \tfrac{1}{2}[\mathrm{AJ}(\mathrm{H},\mathrm{R}) + \mathrm{AJ}(\mathrm{RO},\mathrm{R})],\\
\text{E–P} &= \tfrac{1}{2}[\mathrm{AJ}(\mathrm{H},\mathrm{CF}) + \mathrm{AJ}(\mathrm{RO},\mathrm{CF})],\\
\text{R–P} &= \mathrm{AJ}(\mathrm{R},\mathrm{CF}),
\end{aligned}$$

and $\mathrm{ECS}_{\mathrm{adj}} = \mathrm{mean}(\text{E–R}, \text{E–P}, \text{R–P})$. This paradigm balancing reduces CF's share of the composite from $3/5$ to $1/3$—important because CF is by far the least reliable strategy. The *primary estimand* is complete-case $\mathrm{ECS}_{\mathrm{adj}}$ (all three components defined); the available-component variant is a labelled sensitivity, never the headline, because for over half its rows it reflects a single paradigm pair.

### 3.3 Reliability ceilings and disattenuation

Each elicitation strategy is an *instrument*, and instruments have test–retest reliability. We measure it per model $\times$ dataset $\times$ strategy by re-eliciting every explanation with a semantically equivalent paraphrase of the prompt on a seeded 50-instance cohort, scoring same-strategy agreement on the AJ scale: $\mathrm{rel}_s = \mathrm{AJ}(\mathrm{base}_s, \mathrm{paraphrase}_s)$. Pooled reliabilities are R 0.90, RO 0.75, H 0.64, CF 0.62 (per-cell values in Appendix Table 10); even at temperature 0 with byte-identical prompts, no model exactly reproduced its own salience list in a single-instance retest probe (Jaccard 0.82–0.91), consistent with batched-inference nondeterminism (Thinking Machines Lab, 2025).

Observed between-instrument agreement is attenuated by instrument unreliability; the classical correction (Spearman, 1904) is

$$\mathrm{corr}(A, B) = \mathrm{obs}(A, B) \,/\, \sqrt{\mathrm{rel}_A \cdot \mathrm{rel}_B}. \quad (3)$$

The interpretation was registered in advance: corrected $\approx 1$ means the pair's divergence is within elicitation noise; corrected $\ll 1$ with a CI excluding 1 means real paradigm divergence beyond instrument unreliability. Spearman's formula is defined for product-moment correlations, so applying it to mean agreement levels is an analogue, which we validate with planted-agreement simulations on the AJ scale (Appendix C): disattenuation cuts mean recovery error roughly fourfold, is essentially unbiased when both reliabilities are $\geq 0.60$ (bias $+0.01$)—the regime of every pooled pair below—and acquires an upward bias only near the $0.30$ floor, which is what the floor exists to exclude. Pairs are estimable only if both reliabilities $\geq 0.30$ with ceiling $n \geq 10$; excluded pairs are reported as excluded (in this run, the three CF pairs of one cell, Nova Pro/AG News, where CF paraphrase reliability falls to $0.16$). Values above 1 (sampling error) are reported with an at-ceiling flag, never truncated. CIs come from a seeded cluster bootstrap (B=2,000) resampling instances and ceiling draws independently.

### 3.4 Pre-registration and inference

The analysis plan was written and amended in dated stages, all *before* the production run: the estimator and its validation gates; the estimand–test alignment; the per-dataset co-primary granularity and instrument fixes; and a final amendment adding the reliability correction, the CAD-IMDb arm, and the simulatability bridge. Family (a), primary: per-cell one-sided sign-flip permutation tests (10,000 permutations, seed 42) of complete-case $\mathrm{ECS}_{\mathrm{adj}} > 0$, Holm-corrected (Holm, 1979) across the 12 cells. Family (a2): the same test on available components (sensitivity). Family (b): consensus-core erasure vs. matched random control, Holm across operators. Family (c): the simulatability bridge (§5.5). Everything else is descriptive. The registered plan with its dated amendments is available to reviewers as an anonymized read-only record,[1] and the per-instance artifacts are released with the code.

## 4 Experimental Setup

**Models.** Amazon Nova Pro, Qwen3-235B-A22B-2507, and DeepSeek-V3—two open-weight families (Qwen3, DeepSeek) and one proprietary, API-only model (Nova Pro), all served through AWS Bedrock (temperature 0, fixed decoding parameters). The trio reflects availability constraints, not a principled sample of frontier systems (§7).

**Datasets.** SST-2 (Socher et al., 2013), MNLI (Williams et al., 2018), AG News (Zhang et al., 2015), and CAD-IMDb—the counterfactually *revised* IMDb reviews of Kaushik et al. (2020), in which human editors minimally rewrote reviews to flip the label, breaking single-cue shortcuts. Each dataset is a frozen, hand-vetted, label-balanced 200-instance set curated from the source split (dedup, length stratification, quality filters), with datasheets recording provenance and licenses. CAD-IMDb predates model training cutoffs; it addresses the shortcut prong of the easy-benchmark objection, not memorization (§7).

[1] ANONYMIZED PRE-REGISTRATION: `[OSF view-only link--insert before submission]`. Timestamps for each amendment are visible in the registration history.

**Pipeline.** $3 \times 4 \times 200 = 2{,}400$ instances; per instance: classification, verbalized confidence, and the four explanations (26,545 API requests, 7.1M tokens; 100% instance completion; refusals affected 35 of 2,400 instances, 1.5%). Model accuracy spans 80–98% per cell; explanations are elicited for the model's own prediction regardless of correctness. Strategy-level validity is H 98%, R 97%, RO 99%, and CF 59% (range 8–99% per cell)—CF's minimal-edit constraint frequently fails to produce a verified flip, itself a finding replicating the validity–minimality trade-off: unconstrained "free" rewrites flip reliably (90%) but over-edit (0.32 vs. 0.11 edited-token ratio). Complete cases (all three components defined) number 1,153 of 2,400; missingness is CF-driven and handled by the dual complete/available reporting plus a free-CF sensitivity re-analysis (§5.6).

**Erasure protocol.** For each instance we erase the consensus core CC3 (tokens named by $\geq$3 of 4 strategies; present in 92% of instances, mean size 2.8) by masking or deletion, re-classify with the *same* model, and compare flip rates against a random control matched on content-word type *and occurrence* counts. CF rewrites are additionally verified by a held-out model (96% cross-model flip confirmation).

**Simulatability protocol.** On a seeded 50-instance/dataset/model cohort (600 rows), we generate three explanation-independent perturbations per instance (two seeded random content-word erasures, one deterministic top-frequency deletion), obtain the target model's actual labels on them, then ask a *different* model (round-robin simulator) to predict those labels given the original text, the prediction, and *one* strategy's explanation, versus a no-explanation baseline. $\mathrm{gain}_s = \mathrm{acc}_s - \mathrm{acc}_{\mathrm{baseline}}$.

## 5 Results

### 5.1 Agreement is above chance in every cell

Table 1 shows the primary result. Complete-case $\mathrm{ECS}_{\mathrm{adj}}$ is positive in all 12 cells, from 0.284 (Nova Pro/MNLI) to 0.617 (DeepSeek-V3/SST-2), and every cell survives the Holm-corrected sign-flip test at $p_{\mathrm{Holm}}$=.0012. The pooled mean of 0.432 places the four ways of asking "why" about 40% of the way from chance toward the maximum achievable given each pair's evidence-set sizes. The available-component family (a2) agrees (12/12, all $p_{\mathrm{Holm}}$=.0012). The cell means conceal wide, often bimodal per-instance spread—instances with perfect nesting (AJ $=$ 1) coexist with instances at or below chance in every cell (Figure 1, appendix).

The complete-case and available-component columns diverge most on MNLI (0.28–0.44 vs. 0.48–0.51), and the reason is mechanical rather than substantive. MNLI has the second-lowest counterfactual validity (36–46%), so most available-component rows there lose their CF-bearing components and reduce to E–R alone—the component that sits well above R–P (0.53 vs. 0.31 pooled on this dataset). The available estimate is thus higher precisely *because* it drops the hardest paradigm contrast, which is why we pre-registered the complete-case population as primary and report the wider-$N$ variant only beside it. Where CF validity is high (CAD-IMDb, $\geq$ 92%) the two columns nearly coincide, as expected under this account.

Model and dataset structure is systematic: DeepSeek-V3 is the most self-consistent model in all four datasets (model-pooled 0.52 vs. Qwen3-235B 0.41, Nova Pro 0.34), and the shortcut-resistant CAD-IMDb is the hardest dataset for every model (dataset-pooled 0.38)—when single lexical cues are broken by design, the paradigms drift further apart, exactly where explanations matter most.

### 5.2 Paradigm distance, and what reliability correction reveals

Agreement is not flat across paradigm pairs (Table 2). Observed agreement follows a ladder ordered by paradigm distance—within-instance complete-case component means: E–P 0.615 $>$ E–R 0.395 $>$ R–P 0.286 ($N$=1,153)—with the same-paradigm H–RO reference at 0.596. Remarkably, the extraction–counterfactual pairs agree about as strongly as the two extraction methods agree with each other, *before* any reliability correction.

The disattenuation resolves the ambiguity that a raw ladder leaves open. Correcting each pair by its instruments' self-consistency ceilings, the two extraction–perturbation pairs rise to 0.88 and 0.96—statistically at or near the ceiling (RO–CF's CI includes 1). In other words, *most of the observed extraction$\leftrightarrow$counterfactual divergence is elicitation noise, not paradigm disagreement*: when a minimal edit exists and is found, it touches

| Dataset | Model | Complete-case (primary) $N$ | $\mathrm{ECS}_{\mathrm{adj}}$ | $p_{\mathrm{Holm}}$ | Available (sensitivity) $N$ | $\mathrm{ECS}_{\mathrm{adj}}$ |
|---|---|---|---|---|---|---|
| SST-2 | Nova Pro | 109 | 0.414 | .0012 | 186 | 0.426 |
| SST-2 | Qwen3-235B | 125 | 0.449 | .0012 | 188 | 0.443 |
| SST-2 | DeepSeek-V3 | 143 | 0.617 | .0012 | 197 | 0.618 |
| MNLI | Nova Pro | 37 | 0.284 | .0012 | 182 | 0.475 |
| MNLI | Qwen3-235B | 39 | 0.431 | .0012 | 191 | 0.508 |
| MNLI | DeepSeek-V3 | 51 | 0.441 | .0012 | 193 | 0.501 |
| AG News | Nova Pro | 30 | 0.355 | .0012 | 190 | 0.439 |
| AG News | Qwen3-235B | 14 | 0.427 | .0012 | 195 | 0.489 |
| AG News | DeepSeek-V3 | 80 | 0.503 | .0012 | 194 | 0.544 |
| CAD-IMDb | Nova Pro | 154 | 0.301 | .0012 | 198 | 0.287 |
| CAD-IMDb | Qwen3-235B | 178 | 0.370 | .0012 | 199 | 0.354 |
| CAD-IMDb | DeepSeek-V3 | 193 | 0.465 | .0012 | 200 | 0.459 |
| Pooled | | 1153 | 0.432 | — | 2313 | 0.462 |

Table 1: **Family (a), primary result.** Complete-case $\mathrm{ECS}_{\mathrm{adj}}$ (0 = chance, 1 = size-adjusted ceiling) per model×dataset cell, with the pre-registered one-sided sign-flip test (Holm across 12 cells; all raw $p \leq 2 \times 10^{-4}$). The available-component estimate is the pre-registered wider-$N$ sensitivity family (a2, also 12/12 significant, all $p_{\mathrm{Holm}}$=.0012). Pooled rows are descriptive (instances cluster within models). Complete-case $N$ varies because completeness requires a verified minimal counterfactual; see §5.6 for the missingness analysis.

| Pair | Comp. | Obs. | $\mathrm{rel}_A \times \mathrm{rel}_B$ | Corrected [95% CI] |
|---|---|---|---|---|
| H–RO | (E–E) | 0.596 | — | — |
| RO–CF | E–P | 0.653 | $.75 \times .62$ | 0.956 [0.909, 1.006] |
| H–CF | E–P | 0.555 | $.64 \times .62$ | 0.883 [0.830, 0.935] |
| H–R | E–R | 0.495 | $.64 \times .90$ | 0.654 [0.623, 0.684] |
| RO–R | E–R | 0.424 | $.75 \times .90$ | 0.516 [0.491, 0.539] |
| R–CF | R–P | 0.291 | $.90 \times .62$ | 0.391 [0.352, 0.429] |

Table 2: **Observed vs. reliability-corrected pairwise agreement**, pooled (per-cell values and exclusions in Appendix Table 10). Obs. = mean AJ over instances where the pair is defined; rel = paraphrase self-consistency ceiling of each strategy; Corrected = Spearman-disattenuated agreement. H–RO is the same-paradigm reference (excluded from $\mathrm{ECS}_{\mathrm{adj}}$; no registered correction). Counterfactual pairs reach the ceiling once instrument noise is removed; rationale pairs do not.

the same tokens the extractive paradigms score highest. By contrast, every rationale-involving pair stays far below its ceiling with CIs excluding 1: H–R 0.654, RO–R 0.516, and R–CF 0.391—even though the rationale is the *most* reliable instrument (rel = 0.90), so its divergence cannot be blamed on noisy elicitation. The one-sentence free-text rationale systematically names different evidence than both the extractive and the counterfactual paradigms. A pre-registered soft-matching sensitivity (§5.6) shows this is not lexical variation (synonyms explain ≈0% of the gap): rationales differ *evidentially*.

Two caveats attach to the at-ceiling reading specifically. Reliabilities are single-paraphrase point estimates, and since low reliability *inflates* the corrected ratio, an unusually disruptive CF paraphrase would push values up exactly where this claim lives (more paraphrases are the highest-value extension; §Limitations). And while the recovery simulation bounds the mechanical bias at ≈0.04 for the pooled pairs' reliabilities (all $\geq 0.62$)—too small to manufacture an effect this size—per-cell entries whose CF reliability nears the floor (e.g. DeepSeek-V3/AG News, 0.32) plausibly overshoot, and carry at-ceiling flags for that reason. The rationale-divergence conclusion is exposed to neither caveat: R is the *most* reliable instrument, and its corrected pairs sit 0.35–0.61 below the ceiling.

The pattern is strongest exactly where shortcuts are broken: on CAD-IMDb, corrected R–CF falls to 0.12–0.35 across models while five of the six corrected extraction–CF pairs lie between 0.88 and 1.06 (the exception is Nova Pro's H–CF at 0.55; Appendix Table 10). This is consistent with the necessity/sufficiency distinction of Kommiya Mothilal et al. (2021): attribution-style and counterfactual evidence coincide when one cue is both sufficient and necessary, but free-text rationalization drifts toward plausible narrative rather than the tokens that actually decide the flip.

### 5.3 Explanations are paradigm-shaped, not model-shaped

If each explanation paradigm were a window onto a model-specific reasoning process, one model's

| Dataset | Cross | Within | $\Delta_{\text{matched}}$ [95% CI] |
|---|---|---|---|
| SST-2 | 0.622 | 0.498 | +0.129 [0.073, 0.190] |
| MNLI | 0.596 | 0.495 | +0.046 [0.009, 0.087] |
| AG News | 0.569 | 0.491 | +0.066 [0.035, 0.100] |
| CAD-IMDb | 0.585 | 0.367 | +0.218 [0.187, 0.248] |

Table 3: **Cross-model contrast** (AJ scale). Cross = same strategy, different models, mean AJ; Within = one model's cross-paradigm $\text{ECS}_{\text{adj}}$. $\Delta_{\text{matched}}$ is the pre-registered strategy-matched paired per-instance contrast ({H, R, RO} cross-model pairs vs. the within-model extraction–rationalization component), which controls CF representation; all four CIs exclude 0.

| Model | Op. | CC3 | Random | Gap | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|
| DeepSeek-V3 | mask | 0.316 | 0.123 | 0.192 | .0002 |
| DeepSeek-V3 | delete | 0.330 | 0.125 | 0.205 | .0002 |
| Nova Pro | mask | 0.291 | 0.147 | 0.144 | .0002 |
| Nova Pro | delete | 0.311 | 0.132 | 0.179 | .0002 |
| Qwen3-235B | mask | 0.276 | 0.104 | 0.172 | .0002 |
| Qwen3-235B | delete | 0.305 | 0.112 | 0.193 | .0002 |

Table 4: **Family (b): consensus-core erasure.** Flip rate of the model's own prediction after erasing CC3 tokens (named by $\geq$3 strategies) vs. an occurrence-matched random content-word control; one-sided sign-flip permutation on within-instance paired differences, Holm across operators.

four explanations should cohere more than two different models' renditions of the same paradigm. The opposite holds (Table 3): same-strategy agreement *across* models exceeds cross-paradigm agreement *within* a model on all four datasets, with the strategy-matched paired contrast positive everywhere (from $+0.046$ on MNLI to $+0.218$ on CAD-IMDb). The elicitation paradigm, not model identity, is the dominant factor shaping which evidence an explanation names. Practically: swapping the model changes the explanation less than swapping the question format.

Cross-model pairs are *not* conditioned on the two models having predicted the same label: they agree on 93.9% of the 8,074 pairs, and the remaining 6.1% compare explanations of different predicted labels. This makes the contrast conservative—restricting to label-matched pairs could only raise the cross-model side—so the direction of the finding is unaffected.

### 5.4 Consensus cores are causally load-bearing

Do the tokens the paradigms agree on actually matter to the prediction? Erasing the CC3 consensus core flips the model's own prediction in 28–33% of instances, versus 10–15% for occurrence-matched random controls—a 2.0–2.7$\times$ ratio, significant for every model under both erasure operators (Table 4; Figure 2 in the appendix plots both panels). The gap grows monotonically with agreement tier (pooled, delete operator: $0.152 \to 0.206 \to 0.213$ for low/mid/high-$\text{ECS}_{\text{adj}}$ terciles; descriptive). Following Parcalabescu and Frank (2024), we interpret this strictly as a second *consistency* axis—stated evidence coincides with revealed input sensitivity—not as ground-truth faithfulness: the core is extracted from, and the flip verified by, the same model (Atanasova et al., 2020; DeYoung et al., 2020). The control is also modest in what it rules out: matched-random shows consensus beats *noise*, not that it beats any salient-token heuristic. Per-strategy flip rates give a partial answer—individual evidence sets are *more* disruptive than the consensus core (H 0.44–0.45, CF 0.45–0.49 vs. CC3 0.29–0.32), as expected since they are larger and unfiltered—but a like-sized non-consensus salient baseline (single-strategy or top-TF-IDF tokens) is the sharper test and is not run here.

### 5.5 Agreement does not predict simulatability under generic perturbations

Cross-paradigm agreement would be decision-relevant if it flagged when explanations fail their core job: helping an observer predict model behavior (Chen et al., 2024). One construct note: Chen et al. (2024) derive probe inputs from the explanation under evaluation, whereas our perturbations are deliberately explanation-*independent* so that one perturbation set is fair to all four strategies. We therefore test whether explanations help predict behavior on *generic* corruptions—a weaker claim than explanation-derived simulatability, and why we qualify this subsection's conclusions accordingly. The pre-registered test answers no (Table 5). First, a stark descriptive fact: explanations barely help at all. Simulator accuracy with an explanation differs from the no-explanation baseline by $-0.07$ to $+0.04$; the largest gain (DeepSeek-V3 rationales, $+0.038$) is small, and counterfactuals *hurt* simulation of Qwen3-235B ($-0.069$). Second, the registered association is null: instance-level $\text{ECS}_{\text{adj}}$ does not correlate with simulatability gain for any model (c1: $|\rho| \leq 0.015$, all Holm $p$=1.0; c2 tercile contrasts: all Holm $p \geq 0.99$), and a low-

| Target model | Baseline acc. | Simulatability gain over baseline H | R | CF | RO | $\rho(\mathrm{ECS_{adj}}, \text{gain})$ [95% CI] | $p_{\mathrm{Holm}}$ |
|---|---|---|---|---|---|---|---|
| Nova Pro | 0.877 | −0.009 | −0.002 | −0.006 | −0.001 | −0.011 [−0.155, 0.144] | 1.00 |
| Qwen3-235B | 0.870 | +0.005 | +0.012 | −0.069 | +0.003 | +0.006 [−0.141, 0.172] | 1.00 |
| DeepSeek-V3 | 0.767 | +0.016 | +0.038 | +0.020 | +0.013 | −0.015 [−0.159, 0.146] | 1.00 |
| Pooled | 0.838 | +0.004 | +0.016 | −0.015 | +0.005 | −0.001 [−0.091, 0.088] | — |

Table 5: **Family (c), pre-registered null.** Simulatability on 600 cohort instances under *explanation-independent* perturbations (a deliberately strategy-fair variant of the explanation-derived construct of Chen et al., 2024; see text): a held-out simulator model predicts the target model's labels on perturbed inputs, given one strategy's explanation vs. no explanation. Explanations barely move simulator accuracy, and instance-level $\mathrm{ECS_{adj}}$ does not predict simulatability gain (c1 Spearman shown; the c2 tercile contrasts are likewise null, all Holm $p \geq 0.99$).

agreement "red flag" has precision $\leq$ 0.04 for identifying instances where explanations mislead. High baselines are part of the story (perturbed predictions are guessable from the text alone at 77–88%, leaving limited headroom), but the conclusion within this design is unambiguous: internal consistency among self-explanations is not a proxy for their utility to an observer *on this task family*. Whether agreement predicts the stronger, explanation-derived simulatability construct remains open; testing it requires per-strategy perturbation sets and is the natural follow-up. We report this null as pre-registered; it constrains how our—and others'—agreement metrics should be used.

### 5.6 Robustness

Six pre-registered sensitivity analyses probe the main results. **(i) Missingness:** complete-case membership is CF-gated (a verified minimal edit must exist). Substituting the unconstrained free-CF rewrite for the minimal one yields complete-case $\mathrm{ECS_{adj}}$ 0.382 ($N$=1,318), positive in all cells: conclusions survive removing the minimal-edit gate. **(ii) Chance model:** the uniform null assumes evidence tokens are drawn uniformly from the instance vocabulary, whereas all strategies preferentially select salient-looking words; a frequency-weighted null shifts the pooled estimate by only $-0.011$ ($0.432 \rightarrow 0.421$), with every cell still positive. The shift is small by construction: the urn is the *content-word* vocabulary after stopword removal and lemmatization, so the high-frequency function words that would otherwise dominate a frequency-weighted draw are already absent, leaving a comparatively flat weight distribution over the remaining candidates. **(iii) Lexical variation:** scoring with embedding-based soft matching (pinned static vectors, cosine $\tau$=0.8, matched MC null) leaves complete-case $\mathrm{ECS_{adj}}$ unchanged (0.4320 at $\tau$=0.8) and closes $\approx 0\%$ of the extraction–perturbation vs. rationale-pair gap (+0.200 hard vs. +0.204 soft): the rationale divergence of §5.2 is evidential, not synonymy. **(iv) Estimand family:** the available-component family (a2) reproduces 12/12 significance. **(v) Test conservativeness:** the AJ floor asymmetry makes the one-sided sign-flip test conservative by construction. **(vi) Confidence:** verbalized confidence is degenerate at temperature 0 (mean 0.94 with heavy clustering at 0.95), replicating Xiong et al. (2024); its association with $\mathrm{ECS_{adj}}$ is weak and inconsistent (per-cell Spearman $-0.16$ to $+0.31$; Appendix Table 8) and we draw no conclusion from it. **(vii) Degeneracy exclusions:** the $\varepsilon$ guard drops pairs unevenly (17.3% of H–R and 17.0% of H–CF slots vs. 2.5% of RO–R), so we re-derived the ladder on the 850 complete cases with *no* degenerate pair: E–P $0.618 >$ E–R $0.451 >$ R–P 0.313—same ordering as the full set (0.615/0.395/0.286), so the guard does not manufacture it (Appendix D). **(viii) Prediction correctness:** complete-case $\mathrm{ECS_{adj}}$ is 0.437 on correct predictions ($n$=1,085) vs. 0.360 on incorrect ones ($n$=68), with no consistent per-cell direction (Table 9): agreement is not an artifact of correct instances, though the incorrect stratum is too small for inference.

## 6 Discussion

**Paradigms are complements, not substitutes.** The reliability-corrected picture says the four standard ways of asking "why" collapse into roughly two evidential objects, not one and not four: an extraction/counterfactual cluster agreeing near its noise ceiling, and free-text rationalization, which reliably (in both senses) tells a different story. This cautions against averaging paradigms into one "explanation quality" score, and against treating a rationale as a cheap summary of token-level evidence—on shortcut-broken data the two are barely related (corrected R–P as low as 0.12).

**Measure the instrument before the construct.** Uncorrected, our data would "show" substantial extraction–counterfactual disagreement (observed 0.56–0.65); about three quarters of that apparent shortfall is elicitation noise, which the paraphrase ceiling makes visible and the disattenuation removes. Agreement studies of LLM explanations should routinely report self-consistency ceilings—observed agreement without them conflates two very different failure modes.

**Consistency $\neq$ utility.** The erasure result says consensus evidence is causally load-bearing; the simulatability null says none of this measurably helps an observer predict the model on generic corruptions of the input. Both can be true: our erasure perturbs the tokens explanations name, while simulatability requires explanations to transfer *decision structure*, which one-shot label-conditioned explanations of high-accuracy classifiers apparently do not carry—at least under explanation-independent perturbations. Agreement metrics—ours included—should therefore be read as descriptions of the explanation-generating process, not as certificates of usefulness.

## 7 Conclusion

Four self-explanation paradigms, measured across three LLM families and four datasets under a pre-registered design, agree above chance everywhere—but not uniformly. Once instrument reliability is accounted for, counterfactual and extractive evidence coincide up to elicitation noise while free-text rationales diverge evidentially; the paradigm shapes the explanation more than the model does; consensus tokens are causally load-bearing; and none of this predicts simulatability under explanation-independent perturbations. The measurement stack (exact-null adjusted Jaccard, paradigm balancing, paraphrase ceilings, disattenuation) is model-agnostic and, we hope, useful wherever explanation agreement is quantified.

## Limitations

**Consistency, not faithfulness.** Every axis here—cross-paradigm agreement, erasure, simulatability—is a consistency measure among model outputs (Parcalabescu and Frank, 2024; Jacovi and Goldberg, 2020). High agreement does not certify that any explanation reflects the model's computation; all four paradigms could converge on the same plausible-but-unfaithful cue.

**Label-conditioned elicitation.** All explanation prompts reveal the model's predicted label, so we measure consistency of *post-hoc rationalizations* (Wiegreffe et al., 2021; Turpin et al., 2023). The shared anchor could inflate agreement by pointing every strategy at the same obvious cue; the chance/ceiling adjustment does not remove this, and a no-label control is future work.

**Counterfactual-driven selection.** Complete-case $\mathrm{ECS}_{\mathrm{adj}}$ conditions on a verified minimal counterfactual existing; 1,153/2,400 instances (48%) qualify. Complete and incomplete instances are indistinguishable on every agreement component measurable for both, on length, vocabulary, and accuracy, and the free-CF sensitivity reproduces all conclusions, but selection on unobservables cannot be excluded. Relatedly, the attribution$\leftrightarrow$intervention gap we quantify is partly definitional (Kommiya Mothilal et al., 2021): flip-sets and support-sets need not coincide; our contribution is measuring the size and reliability-adjusted reality of that gap, not adjudicating it as error.

**Models and data.** Three mid-to-large model families—two open-weight, one proprietary—accessed through one provider; none is a current frontier system (an access constraint, not a design choice). All four datasets predate training cutoffs and are plausibly contaminated (Sainz et al., 2023); accuracy is 80–98%, so we describe explanations of largely easy predictions—CAD-IMDb blunts the shortcut concern but not the contamination concern. English classification tasks only; generation and reasoning tasks are untested.

**Instrument scope.** Reliability ceilings come from *one* paraphrase per prompt on 50-instance cohorts, so each $\mathrm{rel}_s$ is a point estimate that conflates the instrument's stability with the particular paraphrase's disruptiveness. This matters asymmetrically: because low reliability inflates the corrected ratio, an unusually disruptive CF paraphrase would bias the extraction$\leftrightarrow$CF "at ceiling" reading upward, and our simulations bound but do not eliminate that concern (§3.3). Two to three additional paraphrases per strategy—roughly 5k further calls—would convert each ceiling into a distribution and let the corrected conclusions be reported across the reliability spread; we regard this as the single highest-value extension of this work. The rationale-divergence result is not exposed to this risk (§5.2). Verbalized confidence is degenerate at temperature 0 and uninformative here. Simulatability used three perturbations per instance and one simulator per target; low power against very small effects cannot be excluded, though the point estimates are near-exactly zero.

## Ethics Statement

All data are public English-language benchmark texts; no human subjects, annotators, or personal data were involved. The study consumed approximately 7.1M tokens of commercial LLM inference. Findings indicate that LLM self-explanations, while internally consistent above chance, are paradigm-dependent and of unproven utility to observers; deploying them as trust signals in high-stakes settings would be premature, and our results should not be cited as evidence that any studied model "explains itself" faithfully.

## Reproducibility Statement

The full pipeline (collection, parsing, normalization, metrics, statistics, figures), the frozen prompt templates with SHA-256 manifest, curated dataset files with datasheets, the dated preregistration with all amendments, per-instance artifacts, and every number in this paper are generated from a single archived run (seed 42, temperature 0, checkpointed and resumable; 600+ unit/property tests, including planted-agreement simulations validating the estimator). Code and artifacts will be released upon publication.

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## A Elicitation Prompts

All strategies are elicited with frozen templates (SHA-256 manifest in the release); the classification and confidence prompts, and dataset-specific CF variants (multiclass label sets, NLI hypothesis editing), appear in the release. The four core templates, abridged to their instruction lines ($\{\cdot\}$ = filled fields; every prompt opens with *"The text was classified as: {predicted label}"* and the text, and requests JSON-only output):

**H (highlighting).** "Analyze the importance of each word for this classification. Give every word an importance score from 1 (irrelevant) to 10 (decisive), using each word exactly as it appears in the text, in order. Include every occurrence, even repeated words."

**R (rationale).** "In one sentence, explain why the text was classified as {predicted label}."

**CF (counterfactual).** "Edit the text so that it is classified as {other label(s)} instead. Make as few edits as possible—change at most a third of the words, and do not add new sentences. Keep the original spacing and punctuation style; only change the words you edit."

**RO (rank ordering).** "Identify the 5 most important words for this classification, ranked from most to least important. Each item must be a single word copied from the text."

Each strategy also has a frozen semantically-equivalent paraphrase used only for the self-consistency ceiling (§3.3).

## B Coverage, Components, and Confidence

Table 6 reports per-strategy validity and complete-case counts per cell. Table 7 reports the per-cell paradigm components (available-component means; the within-instance complete-case comparison is in §5.2). Table 8 reports the verbalized-confidence association, and Table 9 the complete-case estimate split by prediction correctness. Figure 3 shows the component heatmap beside the raw-Jaccard view it replaces; Figure 4 shows the counterfactual validity–minimality trade-off; Figure 5 the confidence–agreement grid.

## C Validating the Disattenuation on the AJ Scale

Spearman's correction is derived for product-moment correlations; we apply it to mean adjusted-Jaccard agreement, so we validate the analogue by simulation rather than assume it.

**Design.** Each instrument observes a latent evidence set $T_s \subseteq V$. One elicitation keeps each token of $T_s$ independently with probability $q_s$ and replaces each dropped token with a uniform random distractor—size-preserving, matching AJ's fixed-size geometry. For a configuration $(V, a, b, \text{overlap}, q_A, q_B)$ we draw 2,000 elicitation pairs and compute $\text{rel}_s$ as mean AJ between two independent elicitations of the same $T_s$, obs as mean AJ across instruments, and $\text{true} = \text{AJ}(T_A, T_B)$ noise-free. The grid crosses four geometries (balanced, imbalanced, large-$V$, small-$V$), $q \in \{0.55, 0.7, 0.85, 1.0\}$, and three planted overlap levels: 120 configurations, 66 above the pre-registered floor.

**Result.** Above the floor, disattenuation recovers the planted value with mean $|\text{corrected} - \text{true}| = 0.050$, against mean $|\text{obs} - \text{true}| = 0.260$ on the noisy configurations—a fourfold error reduction. Bias depends on reliability: in the band $\min(\text{rel}) \geq 0.60$ recovery is essentially unbiased (signed bias $+0.009$, max error $0.041$), in $[0.45, 0.60)$ bias is $+0.047$, and in $[0.30, 0.45)$ it reaches $+0.072$ (max $+0.156$, on configurations with planted agreement $1.0$). Below the floor, maximum error grows to $0.245$. Two consequences for the paper: every pooled pair in Table 2 sits in the unbiased regime (lowest pooled reliability is CF at 0.62), and the upward near-floor bias means low-reliability *per-cell* entries should be read as upper bounds—which is what the at-ceiling flag marks. These assertions are pinned in the released test suite, in `tests/` as `test_disattenuation_recovery.py`.

## D Degeneracy Exclusions by Pair Type

The $\varepsilon = 0.10$ guard removes pairs whose geometry puts the ceiling within $\varepsilon$ of the chance floor. Exclusion rates differ sharply by pair type: H–R 17.3% of elicited slots (396/2286), H–CF 17.0% (238/1397), R–CF 13.7% (189/1375), RO–CF 4.8% (67/1403), RO–R 2.5% (58/2304). RO's fixed 5-token evidence sets rarely produce degenerate geometry, whereas H's length-proportional sets shrink to 3 tokens on short inputs, where $J_{\max}$ and $\mathbb{E}[J]$ converge. Exclusions concentrate on the short-input datasets (CAD-IMDb 398 pair-slots, MNLI 289, SST-2 213, AG News 48).

Because the guard falls hardest on H- and CF-involving pairs, it could in principle shape the ladder. It does not: restricting to the 850 complete cases containing *no* degenerate pair reproduces the ordering and the separations (E–P 0.618, E–R 0.451, R–P 0.313; full complete-case set 0.615, 0.395, 0.286). The E–R component rises slightly under the restriction, which if anything narrows the E–P/E–R gap while leaving R–P clearly lowest.

## E Disattenuated Agreement, Per Cell

Table 10 reports, for every cell and pair: observed AJ, the two reliabilities, the corrected value with bootstrap CI, at-ceiling flags, and the pre-registered floor exclusions (CF on Nova Pro/AG News, where CF paraphrase reliability is 0.16).

Table 6: Per-strategy extraction success and complete-case counts, per model×dataset.

| Cell | N | H | R | CF | RO | Complete |
|---|---|---|---|---|---|---|
| DeepSeek-V3/AG News | 200 | 1.000 | 0.960 | 0.415 | 1.000 | 81 |
| DeepSeek-V3/CAD-IMDb | 200 | 1.000 | 1.000 | 0.985 | 1.000 | 197 |
| DeepSeek-V3/MNLI | 200 | 1.000 | 0.965 | 0.460 | 0.965 | 88 |
| DeepSeek-V3/SST-2 | 200 | 1.000 | 0.955 | 0.905 | 0.990 | 172 |
| Nova Pro/AG News | 200 | 0.995 | 0.950 | 0.155 | 1.000 | 31 |
| Nova Pro/CAD-IMDb | 200 | 0.965 | 0.985 | 0.920 | 0.995 | 177 |
| Nova Pro/MNLI | 200 | 0.995 | 0.940 | 0.355 | 0.970 | 65 |
| Nova Pro/SST-2 | 200 | 0.990 | 0.960 | 0.650 | 0.990 | 124 |
| Qwen3-235B/AG News | 200 | 0.975 | 0.975 | 0.075 | 1.000 | 15 |
| Qwen3-235B/CAD-IMDb | 200 | 0.955 | 0.990 | 0.930 | 1.000 | 179 |
| Qwen3-235B/MNLI | 200 | 0.945 | 0.990 | 0.400 | 0.975 | 77 |
| Qwen3-235B/SST-2 | 200 | 1.000 | 0.940 | 0.810 | 0.985 | 149 |

Table 7: Paradigm-component adjusted Jaccard: E–R, E–P, R–P per cell.

| Cell | E–R | E–P | R–P |
|---|---|---|---|
| DeepSeek-V3/AG News | 0.556 | 0.522 | 0.456 |
| DeepSeek-V3/CAD-IMDb | 0.403 | 0.704 | 0.277 |
| DeepSeek-V3/MNLI | 0.526 | 0.501 | 0.367 |
| DeepSeek-V3/SST-2 | 0.522 | 0.770 | 0.581 |
| Nova Pro/AG News | 0.446 | 0.376 | 0.275 |
| Nova Pro/CAD-IMDb | 0.233 | 0.531 | 0.090 |
| Nova Pro/MNLI | 0.531 | 0.248 | 0.191 |
| Nova Pro/SST-2 | 0.442 | 0.485 | 0.331 |
| Qwen3-235B/AG News | 0.490 | 0.444 | 0.410 |
| Qwen3-235B/CAD-IMDb | 0.251 | 0.703 | 0.138 |
| Qwen3-235B/MNLI | 0.543 | 0.477 | 0.338 |
| Qwen3-235B/SST-2 | 0.339 | 0.750 | 0.258 |

Table 8: Confidence↔ECS association (Spearman $\rho$, Kendall $\tau_b$), per cell.

| Cell | $\rho$ | $\tau_b$ | $p$ | N |
|---|---|---|---|---|
| DeepSeek-V3/AG News | -0.113 | -0.090 | 0.1177 | 194 |
| DeepSeek-V3/CAD-IMDb | 0.310 | 0.235 | 0.0000 | 200 |
| DeepSeek-V3/MNLI | 0.026 | 0.020 | 0.7172 | 192 |
| DeepSeek-V3/SST-2 | 0.215 | 0.169 | 0.0024 | 198 |
| Nova Pro/AG News | -0.164 | -0.130 | 0.0244 | 189 |
| Nova Pro/CAD-IMDb | 0.079 | 0.062 | 0.2702 | 197 |
| Nova Pro/MNLI | 0.207 | 0.156 | 0.0046 | 187 |
| Nova Pro/SST-2 | 0.209 | 0.164 | 0.0034 | 194 |
| Qwen3-235B/AG News | -0.009 | -0.005 | 0.9065 | 190 |
| Qwen3-235B/CAD-IMDb | 0.146 | 0.112 | 0.0410 | 196 |
| Qwen3-235B/MNLI | 0.122 | 0.092 | 0.0973 | 186 |
| Qwen3-235B/SST-2 | 0.175 | 0.137 | 0.0137 | 198 |

| | Correct | | Incorrect | |
|---|---|---|---|---|
| Cell | $\mathrm{ECS}_{\mathrm{adj}}$ | $N$ | $\mathrm{ECS}_{\mathrm{adj}}$ | $N$ |
| DeepSeek-V3/AG News | 0.505 | 70 | 0.483 | 10 |
| DeepSeek-V3/CAD-IMDb | 0.475 | 187 | 0.154 | 6 |
| DeepSeek-V3/MNLI | 0.463 | 40 | 0.360 | 11 |
| DeepSeek-V3/SST-2 | 0.615 | 140 | 0.705 | 3 |
| Nova Pro/AG News | 0.350 | 27 | 0.398 | 3 |
| Nova Pro/CAD-IMDb | 0.306 | 145 | 0.222 | 9 |
| Nova Pro/MNLI | 0.297 | 33 | 0.171 | 4 |
| Nova Pro/SST-2 | 0.406 | 102 | 0.530 | 7 |
| Qwen3-235B/AG News | 0.427 | 14 | — | 0 |
| Qwen3-235B/CAD-IMDb | 0.372 | 170 | 0.345 | 8 |
| Qwen3-235B/MNLI | 0.469 | 35 | 0.095 | 4 |
| Qwen3-235B/SST-2 | 0.444 | 122 | 0.646 | 3 |
| Pooled | 0.437 | 1085 | 0.360 | 68 |

Table 9: Complete-case $\mathrm{ECS}_{\mathrm{adj}}$ split by whether the model's prediction was correct. Descriptive only: the incorrect stratum is small (68 of 1,153 complete cases) and the per-cell direction is inconsistent (agreement is higher on incorrect predictions in 4 of 11 estimable cells).

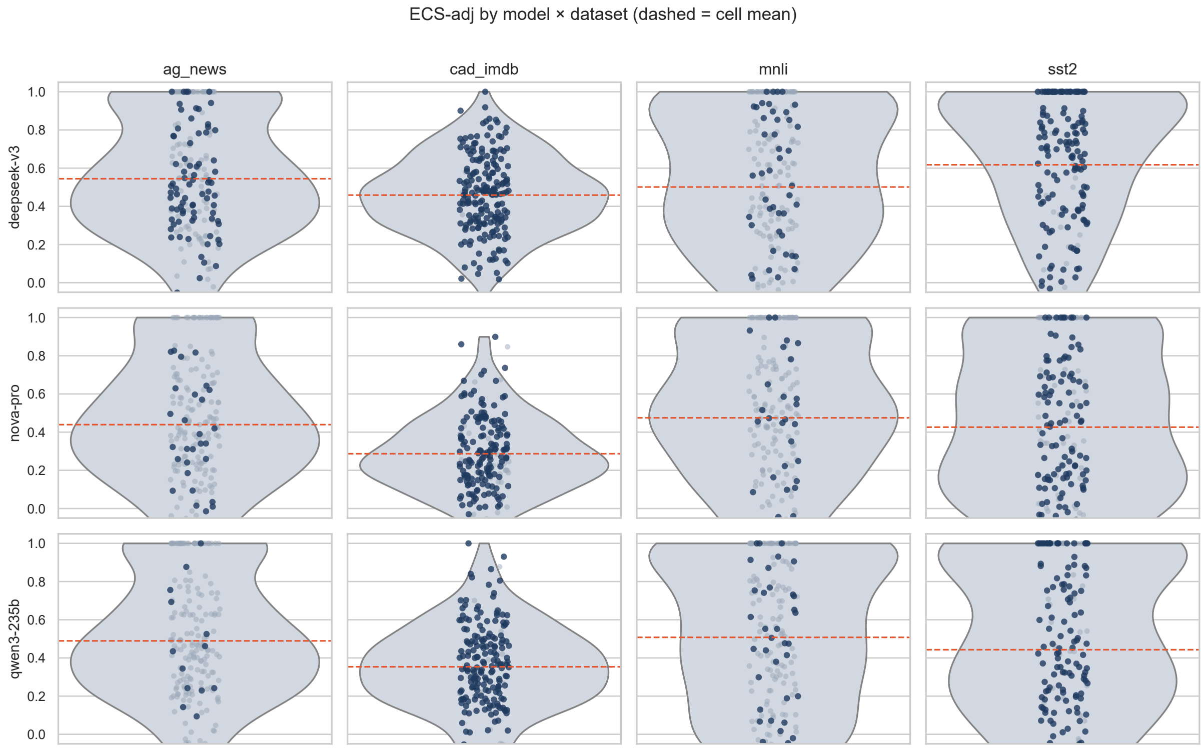


Figure 1: Per-instance $\mathrm{ECS}_{\mathrm{adj}}$ distributions by model (rows) and dataset (columns); dashed line = cell mean; dark points = complete-case instances. All 12 cell means are above chance (Table 1), but distributions are wide and often bimodal.

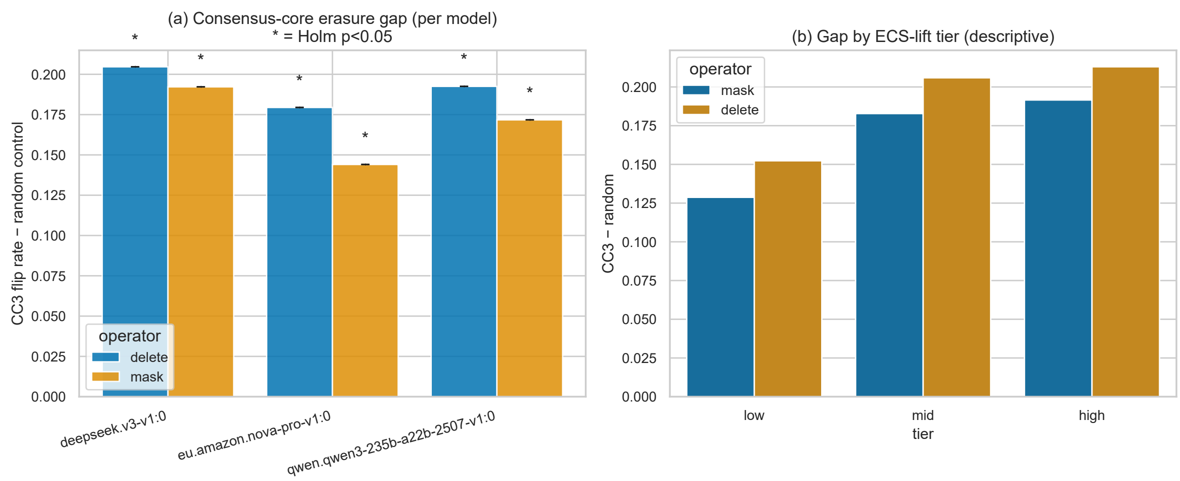


Figure 2: Consensus-core erasure. (a) CC3-minus-random flip-rate gap per model and operator (all Holm $p < .05$). (b) Pooled gap by $\mathrm{ECS}_{\mathrm{adj}}$-lift tier (descriptive): higher cross-paradigm agreement coincides with a larger causal-sensitivity gap.

Table 10: Disattenuated cross-paradigm agreement (Spearman 1904): observed AJ corrected by the geometric mean of the two strategies' self-consistency ceilings, per cell. Pairs below the pre-registered reliability floor (0.30, ceiling $n \geq 10$) are reported as excluded.

| Cell | Pair | Obs. | $rel_A \times rel_B$ | Corrected [95% CI] | Flag |
|---|---|---|---|---|---|
| DeepSeek-V3/AG News | H–R | 0.534 | 0.730×0.910 | 0.655 [0.571, 0.738] | |
| DeepSeek-V3/AG News | RO–R | 0.548 | 0.727×0.910 | 0.674 [0.599, 0.757] | |
| DeepSeek-V3/AG News | H–CF | 0.569 | 0.730×0.316 | 1.185 [0.940, 1.569] | at ceiling |
| DeepSeek-V3/AG News | RO–CF | 0.479 | 0.727×0.316 | 1.000 [0.776, 1.343] | at ceiling |
| DeepSeek-V3/AG News | R–CF | 0.456 | 0.910×0.316 | 0.851 [0.638, 1.178] | |
| DeepSeek-V3/CAD-IMDb | H–R | 0.631 | 0.603×0.762 | 0.931 [0.830, 1.045] | |
| DeepSeek-V3/CAD-IMDb | RO–R | 0.345 | 0.847×0.762 | 0.430 [0.372, 0.495] | |
| DeepSeek-V3/CAD-IMDb | H–CF | 0.616 | 0.603×0.809 | 0.881 [0.804, 0.962] | |
| DeepSeek-V3/CAD-IMDb | RO–CF | 0.797 | 0.847×0.809 | 0.962 [0.901, 1.032] | |
| DeepSeek-V3/CAD-IMDb | R–CF | 0.277 | 0.762×0.809 | 0.353 [0.300, 0.406] | |
| DeepSeek-V3/MNLI | H–R | 0.495 | 0.615×0.950 | 0.647 [0.543, 0.760] | |
| DeepSeek-V3/MNLI | RO–R | 0.551 | 0.750×0.950 | 0.653 [0.568, 0.751] | |
| DeepSeek-V3/MNLI | H–CF | 0.338 | 0.615×0.648 | 0.535 [0.163, 0.915] | |
| DeepSeek-V3/MNLI | RO–CF | 0.546 | 0.750×0.648 | 0.783 [0.569, 1.021] | |
| DeepSeek-V3/MNLI | R–CF | 0.367 | 0.950×0.648 | 0.468 [0.248, 0.721] | |
| DeepSeek-V3/SST-2 | H–R | 0.498 | 0.728×0.957 | 0.596 [0.498, 0.708] | |
| DeepSeek-V3/SST-2 | RO–R | 0.547 | 0.829×0.957 | 0.614 [0.521, 0.711] | |
| DeepSeek-V3/SST-2 | H–CF | 0.753 | 0.728×0.679 | 1.071 [0.927, 1.244] | at ceiling |
| DeepSeek-V3/SST-2 | RO–CF | 0.778 | 0.829×0.679 | 1.037 [0.913, 1.196] | at ceiling |
| DeepSeek-V3/SST-2 | R–CF | 0.581 | 0.957×0.679 | 0.721 [0.596, 0.854] | |
| Nova Pro/AG News | H–R | 0.443 | 0.547×0.922 | 0.624 [0.535, 0.718] | |
| Nova Pro/AG News | RO–R | 0.450 | 0.621×0.922 | 0.594 [0.511, 0.682] | |
| Nova Pro/AG News | H–CF | 0.379 | 0.547×0.164 | – | excl.: rel. floor (CF) |
| Nova Pro/AG News | RO–CF | 0.332 | 0.621×0.164 | – | excl.: rel. floor (CF) |
| Nova Pro/AG News | R–CF | 0.275 | 0.922×0.164 | – | excl.: rel. floor (CF) |
| Nova Pro/CAD-IMDb | H–R | 0.444 | 0.464×0.804 | 0.728 [0.611, 0.856] | |
| Nova Pro/CAD-IMDb | RO–R | 0.201 | 0.680×0.804 | 0.272 [0.217, 0.334] | |
| Nova Pro/CAD-IMDb | H–CF | 0.318 | 0.464×0.717 | 0.552 [0.476, 0.640] | |
| Nova Pro/CAD-IMDb | RO–CF | 0.740 | 0.680×0.717 | 1.060 [0.967, 1.173] | at ceiling |
| Nova Pro/CAD-IMDb | R–CF | 0.090 | 0.804×0.717 | 0.119 [0.065, 0.174] | |
| Nova Pro/MNLI | H–R | 0.529 | 0.664×0.922 | 0.676 [0.572, 0.787] | |
| Nova Pro/MNLI | RO–R | 0.536 | 0.775×0.922 | 0.635 [0.548, 0.729] | |
| Nova Pro/MNLI | H–CF | 0.330 | 0.664×0.346 | 0.690 [0.230, 1.217] | |
| Nova Pro/MNLI | RO–CF | 0.171 | 0.775×0.346 | 0.331 [0.007, 0.723] | |
| Nova Pro/MNLI | R–CF | 0.191 | 0.922×0.346 | 0.338 [-0.028, 0.756] | |
| Nova Pro/SST-2 | H–R | 0.466 | 0.533×0.903 | 0.672 [0.551, 0.813] | |
| Nova Pro/SST-2 | RO–R | 0.415 | 0.698×0.903 | 0.522 [0.407, 0.655] | |
| Nova Pro/SST-2 | H–CF | 0.431 | 0.533×0.698 | 0.706 [0.541, 0.882] | |
| Nova Pro/SST-2 | RO–CF | 0.541 | 0.698×0.698 | 0.775 [0.637, 0.931] | |
| Nova Pro/SST-2 | R–CF | 0.331 | 0.903×0.698 | 0.417 [0.271, 0.558] | |
| Qwen3-235B/AG News | H–R | 0.487 | 0.700×0.943 | 0.600 [0.526, 0.678] | |
| Qwen3-235B/AG News | RO–R | 0.480 | 0.804×0.943 | 0.552 [0.492, 0.618] | |
| Qwen3-235B/AG News | H–CF | 0.465 | 0.700×0.736 | 0.649 [0.324, 1.034] | |
| Qwen3-235B/AG News | RO–CF | 0.434 | 0.804×0.736 | 0.565 [0.272, 0.909] | |
| Qwen3-235B/AG News | R–CF | 0.410 | 0.943×0.736 | 0.492 [0.169, 0.867] | |
| Qwen3-235B/CAD-IMDb | H–R | 0.622 | 0.600×0.902 | 0.846 [0.746, 0.944] | |
| Qwen3-235B/CAD-IMDb | RO–R | 0.163 | 0.816×0.902 | 0.189 [0.145, 0.236] | |
| Qwen3-235B/CAD-IMDb | H–CF | 0.647 | 0.600×0.797 | 0.935 [0.858, 1.018] | |
| Qwen3-235B/CAD-IMDb | RO–CF | 0.752 | 0.816×0.797 | 0.932 [0.864, 1.014] | |
| Qwen3-235B/CAD-IMDb | R–CF | 0.138 | 0.902×0.797 | 0.163 [0.117, 0.212] | |
| Qwen3-235B/MNLI | H–R | 0.549 | 0.735×0.963 | 0.653 [0.565, 0.745] | |
| Qwen3-235B/MNLI | RO–R | 0.533 | 0.729×0.963 | 0.637 [0.543, 0.738] | |
| Qwen3-235B/MNLI | H–CF | 0.506 | 0.735×0.630 | 0.744 [0.424, 1.081] | |
| Qwen3-235B/MNLI | RO–CF | 0.427 | 0.729×0.630 | 0.630 [0.386, 0.886] | |
| Qwen3-235B/MNLI | R–CF | 0.338 | 0.963×0.630 | 0.435 [0.151, 0.720] | |
| Qwen3-235B/SST-2 | H–R | 0.319 | 0.720×0.849 | 0.408 [0.293, 0.530] | |
| Qwen3-235B/SST-2 | RO–R | 0.359 | 0.762×0.849 | 0.446 [0.333, 0.565] | |
| Qwen3-235B/SST-2 | H–CF | 0.744 | 0.720×0.790 | 0.987 [0.861, 1.133] | |
| Qwen3-235B/SST-2 | RO–CF | 0.751 | 0.762×0.790 | 0.967 [0.856, 1.096] | |
| Qwen3-235B/SST-2 | R–CF | 0.258 | 0.849×0.790 | 0.315 [0.186, 0.443] | |

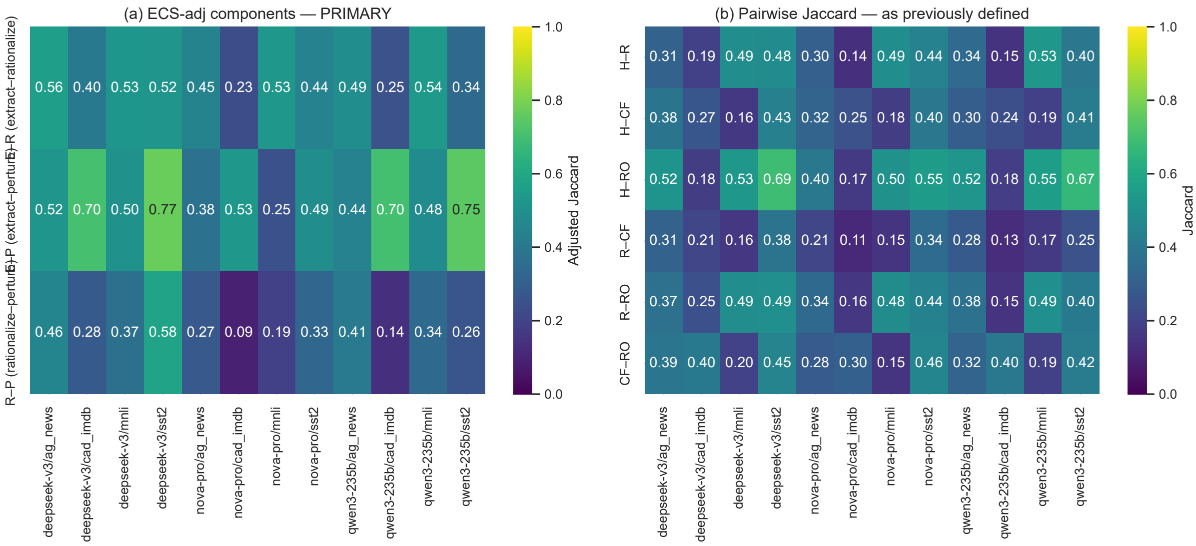


Figure 3: (a) $\mathrm{ECS}_{\mathrm{adj}}$ paradigm components per cell (primary scale). (b) The raw pairwise Jaccard view of the same data (deprecated comparator): without chance/ceiling adjustment, set-size geometry dominates the picture.

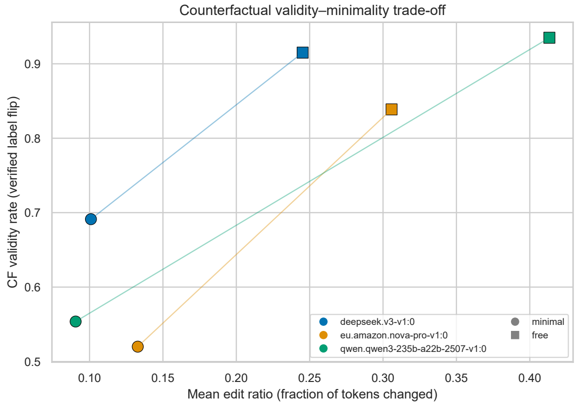


Figure 4: Counterfactual validity–minimality trade-off: minimal-edit CFs flip less often but edit far less; free rewrites flip reliably (90%) but over-edit.

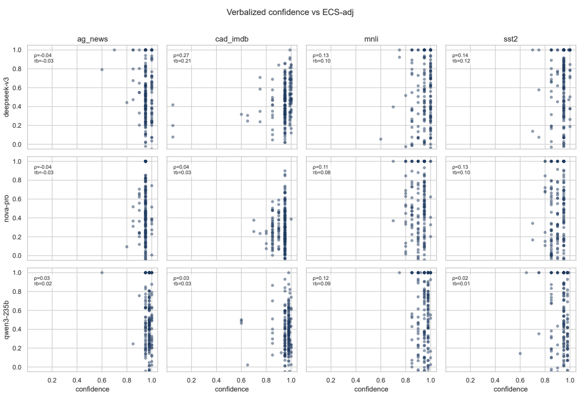


Figure 5: Verbalized confidence vs. $ECS_{adj}$ per cell. Confidence is degenerate at temperature 0 (clustered at 0.95), consistent with Xiong et al. (2024); associations are weak and inconsistent.